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SYSTEMS AND METHODS FOR AUGMENTED REALITY BASED WHOLE SLIDE IMAGE VISUALIZATION

Abstract

A computer-implemented method for generating a navigable three-dimensional image of a tissue sample using augmented reality or virtual reality (AR/VR), the method comprising: receiving, by one or more processors, a plurality of medical images, the plurality of medical images being associated with a whole slide image (WSI) the tissue sample; determining by one or more processors, a layer level of each medical image of the plurality of medical images; determining, by one or more processors, for each medical image of the plurality of medical images, one or more regions of tissue and one or more regions of non-tissue; stacking, by one or more processors, each medical image with one or more of the plurality of medical images based on the layer level to generate a three-dimensional rendering for display in an AR/VR environment; and outputting, by one or more processors, the AR/VR environment as seen in a display.

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Background/Summary

TECHNICAL FIELD

[0001] Various embodiments of the present disclosure pertain generally to pathology slide analysis and related methods. More specifically, particular embodiments of the present disclosure relate to systems and methods for using Augmented Reality/Virtual Reality (AR/VR) for analyzing pathology slides. The present disclosure further provides systems and methods relating to volume rendering.

BACKGROUND

[0002] Pathology specimens may be cut into multiple sections, stained, and prepared as slides for a pathologist to examine and render a diagnosis. Pathology slides, using digital pathology techniques, may be conceptualized and/or prepared as two-dimensional images that represent a cross-section from a larger, three-dimensional sample (e.g., piece of tissue). However, some constraints presented by limiting a perception of slides to two dimensions has translated to digital pathology, where scanned slides (e.g., whole slide images or WSIs) may be presented independently from one another. Such a two-dimensional, piecemeal presentation may make assessing or visualizing the often irregular, three-dimensional shapes and dimensions of biological structures (e.g., tumors) difficult. A desire exists for a way to enable pathologists to see and analyze the depth of the tissue.

[0003] The background description provided herein is for the purpose of generally presenting the context of the disclosure. Unless otherwise indicated herein, the materials described in this section are not prior art to the claims in this application and are not admitted to be prior art, or suggestions of the prior art, by inclusion in this section.

[0004] Additional objects and advantages of the disclosed aspects will be set forth in part in the description that follows, and in part will be apparent from the description, or may be learned by practice of the disclosed aspects. The objects and advantages of the disclosed aspects will be realized and attained by means of the elements and combinations particularly pointed out in the appended claims.

[0005] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the disclosed aspects, as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various exemplary aspects and together with the description, serve to explain the principles of the disclosed aspects.

[0007] FIG. 1 depicts a network environment that may execute techniques presented herein.

[0008] FIG. 2 depicts a flowchart of an exemplary method for generating a navigable three-dimensional image of a tissue sample in an AR/VR environment, according to one or more embodiments.

[0009] FIG. 3 depicts a flowchart of an exemplary method for generating a navigable three-dimensional image of a tissue sample, according to an alternative embodiment.

[0010] FIG. 4 depicts a flowchart of an exemplary method for navigating a three-dimensional image in an AR/VR environment.

[0011] FIG. 5 depicts an example of a computing device, according to one or more embodiments

[0012] Notably, for simplicity and clarity of illustration, certain aspects of the figures depict the general configuration of the various embodiments. Descriptions and details of well-known features and techniques may be omitted to avoid unnecessarily obscuring other features. Elements in the figures are not necessarily drawn to scale; the dimensions of some features may be exaggerated relative to other elements to improve understanding of the example embodiments.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0013] The use of AR/VR technology in electronic image analysis in digital pathology may aid pathologists and other practitioners in making more accurate and timely diagnoses.

[0014] Biological structures (e.g. tumors) may be irregular, three-dimensional shapes. Aspects disclosed herein may take these three-dimensional structures into account to present new opportunities for understanding pathology and treating or addressing pathological illnesses such as cancer.

[0015] One aspect of the invention may involve generating a navigable three-dimensional image of a tissue sample may include receiving a plurality of whole slide images (WSI) associated with the tissue sample. The method may further include providing the plurality of whole slide images to a machine-learning model. The machine-learning model may have been trained, using one or more prior patient and/or synthetically generated sets of whole slide images, to identify one or more positional features within the plurality of whole slide images and output a plurality of relative positional relationships corresponding to each of the plurality of whole slide images. The method may further include generating the navigable three-dimensional image of the tissue sample based on the plurality of relative positional relationships. The method may further include generating an interactive display incorporating the navigable three-dimensional image.

[0016] The method mentioned above may further include providing, to a user interface, the interactive display. In some embodiments, the interactive display occurs in an AR/VR space. Augmented reality (AR) creates an interactive experience for the user, allowing the merging of the physical surroundings with computer generated content. On the other hand, Virtual Reality (VR) creates a fully immersive virtual world for the user.

[0017] As discussed herein, one or more machine-learning models may be trained to understand positional features in image data. Accordingly, machine-learning models disclosed herein are image processing machine-learning models. Such image-processing machine learning models may be trained using image and/or medical related data (e.g., whole slide images, patient data, etc., as discussed herein). An image processing machine-learning model trained to understand (e.g., identify) positional features and/or other image features based on image data may be trained to adjust one or more weights, layers, nodes, biases, and/or synapses based on the image related data. “Positional features” shall be used herein to indicate various visual morphology of a slide image. In examples, positional features may include anatomical features (e.g., cells, subcellular components, interstitial tissue, and the like) and/or histological features (e.g., staining patterns, tissue folds, and the like). In various embodiments, the positional features may also include general distinct landmarks of the gross (e.g., whole) image such as unique curvature, boundaries of tissue versus empty space, or the like. In various embodiments, the positional features may be included within training data for a machine-learning model or may be learned using weak supervision. An image-processing machine-learning model may include components (e.g., weights, layers, nodes, biases, and/or synapses) that collectively associate one or more of: a whole slide image with a depth of a tissue sample, a whole slide image with a position in a tissue sample, a relative positional relationship between two or more whole slide images with the two or more whole slide images; and/or the like. “Relative positional relationship” shall be used herein to indicate a position of two images within a three-dimensional whole. For example, for two slide images that are from separate

samples of a same piece of paraffin-embedded tissue, a relative positional relationship describes the relationship of the two images in three-dimensional space. An image processing machine-learning model may correlate image information and patient medical data in a diagnostic context. An image processing machine-learning model may be trained to adjust one or more weights, layers, nodes, biases, and/or synapses to associate certain image data in view of a diagnostic context. For example, particular image features may be correlated with a particular diagnosis. In another example, particular positional features identified in the image data may be correlated with an inferred and/or determined tissue sample position associated with a whole image slide. In such examples, a three-dimensional image of a tissue sample may be constructed and/or generated from a set of individual whole slide images of the tissue sample by identifying the positional features of each whole slide image and relating the positional features identified for each whole slide image to all others in the set. In various embodiments, the three-dimensional image may be generated using one or more algorithms (e.g., machine-learning algorithms, or the like) to determine an orientation of the slide images relative to one another within three-dimensional space. Then, a second algorithm may be used to infer/fill in blank space between the pieces of tissue (e.g., creating voxels). In further embodiments, and in a case where the images or tissue slices are contorted/distorted, or the like, one or more algorithms may be used to register the images in three-dimensional space. In examples, if this is unsuccessful, the system may output an indication of an error (e.g., an error message or the like).

[0018] One aspect of the invention involves a user immersing themselves in an AR/VR space. A user might be a pathologist attempting to analyze WSIs. Prior to the method discussed in this invention, pathologists who view WSIs may use a standard computer mouse to navigate the image. In one aspect of the invention, a pathologist who uses an AR/VR device to view the image could navigate an image using hand gestures. However, pathologists who use hand gestures experience physical discomfort and fatigue. Other techniques for traversing an AR/VR space as discussed below.

[0019] One aspect of the invention further discussed below involves techniques in which multiple users engages with a shared AR/VR space.

[0020] Two dimensional (2D) images may be associated/co-registered and reconstructed in a 3D rendition. Samples may be scanned, for example via laser, to create 3D renderings of each layer. Each layer of a sample may then be virtually stacked in a Z-dimension. In this way, Z stack images may be used to create volumetric rendition of whole slide images. Pre-processing steps may be implemented prior to input, such as modifying the shape of each layer to match the shape of corresponding layers. This may result in layers stacking evenly such that, when stacked, the original 3D sample is substantially reconstructed. Sections cut from the same section consecutively may be used to create the z-stacks. Aspects of the invention disclosed below envision using an AR/VR space to aid in co-registration of images.

[0021] Upon co-registering WSIs, pathologists can be aided by an AI or Machine Learning (ML) model which will suggest a 3D sample. In this process, the ML model might be trained to determine a missing layer. Aspect of the invention disclosed below elaborate on the using an AR/VR space to suggest a missing layer.

[0022] As used herein, a “machine-learning model” generally encompasses instructions, data, and/or a model configured to receive input, and apply one or more of a weight, bias, classification, or analysis on the input to generate an output. The output may include, for example, a classification of the input, an analysis based on the input, a design, process, prediction, or recommendation associated with the input, or any other suitable type of output. A machine-learning model is generally trained using training data, e.g., experiential data and/or samples of input data, which are fed into the model in order to establish, tune, or modify one or more aspects of the model, e.g., the weights, biases, criteria for forming classifications or clusters, or the like. Aspects of a machine-learning model may operate on an input linearly, in parallel, via a network (e.g., a neural network),

or via any suitable configuration.

[0023] The execution of the machine-learning model may include deployment of one or more machine-learning techniques, such as a transformer model, graph neural network (GNN), linear regression, logistic regression, random forest, gradient boosted machine (GBM), deep learning, and/or a deep neural network. Supervised and/or unsupervised training may be employed. For example, supervised learning may include providing training data and labels corresponding to the training data, e.g., as ground truth. Unsupervised approaches may include clustering, classification or the like. K-means clustering or K-Nearest Neighbors may also be used, which may be supervised or unsupervised. Combinations of K-Nearest Neighbors and an unsupervised cluster technique may also be used. Any suitable type of training may be used, e.g., stochastic, gradient boosted, random seeded, recursive, epoch or batch-based, etc.

[0024] While several of the examples herein involve certain types of machine-learning and artificial intelligence, it should be understood that techniques according to this disclosure may be adapted to any suitable type of machine-learning and/or artificial intelligence. It should also be understood that the examples above are illustrative only. The techniques and technologies of this disclosure may be adapted to any suitable activity.

[0025] While various aspects relating to medical imaging and medical diagnostics (e.g., diagnosis of a medical condition based on medical imaging) are described in the present aspects as illustrative examples, the present aspects are not limited to such examples. For example, the present aspects can be implemented for other types of image processing.

[0026] Techniques described in the current disclosure may utilize systems and methods described in U.S. Provisional App. No. 63/563,031, U.S. application Ser. No. 17/107,433, U.S. application Ser. No. 17/126,596, U.S. application Ser. No. 17/313,617, U.S. application Ser. No. 17/732,857, U.S. application Ser. No. 18/643,400, and U.S. application Ser. No. 17/313,617, all of which are incorporated herein by reference.

[0027] FIG. 1 depicts an exemplary environment **100** that may be utilized with techniques presented herein. One or more user device(s) **112** may communicate across an electronic network **110**. The one or more user device(s) **112** may be associated with a user, e.g., a user that is viewing and/or interacting with a generated navigable three-dimensional image, an administrator of one or more components of environment **100**, and/or the like. As will be discussed in further detail below, one or more computing system(s) **102** may communicate with one or more of the other components of the environment **100** across electronic network **110**.

[0028] The user device(s) **112** may be configured to enable a user to access and/or interact with other systems in the environment **100**. For example, the user device(s) **112** may each be a computer system such as, for example, a desktop computer, a mobile device, a tablet, an augmented/virtual/extended reality device, and etc. In some embodiments, the user device(s) **112** may include one or more electronic application(s), e.g., a program, plugin, browser extension, etc., installed on a memory of the user device(s) **112**. In some embodiments, the electronic application(s) may be associated with one or more of the other components in the environment **100**. For example, the electronic application(s) may include one or more of system control software, system monitoring software, software development tools, etc.

[0029] In various embodiments, the environment **100** may include a data store **114** (e.g., database). The data store **114** may include a server system and/or a data storage system such as computer-readable memory such as a hard drive, flash drive, disk, etc. In some embodiments, the data store **114** includes and/or interacts with an application programming interface for exchanging data to other systems, e.g., one or more of the other components of the environment. The data store **114** may include and/or act as a repository or source for storing image data, whole slide images (WSI), a generated three-dimensional image, patient data, output data (e.g., from a machine-learning model), and the like (e.g., to be provided/transmitted to user device **112** or to/from any of the other components of environment **100**).

[0030] In some embodiments, the components of the environment **100** are associated with a common entity, e.g., a service provider, an account provider, or the like. For example, in some embodiments, computing system **102**, data store **114**, and medical computing system **116** may be associated with a common entity. In some embodiments, one or more of the components of the environment are associated with a different entity than another. For example, computing system **102** may be associated with a first entity (e.g., a service provider) while medical computing system **116** may be associated with a second entity (e.g., a medical institution or provider). The systems and devices of the environment **100** may communicate in any arrangement. As will be discussed herein, systems and/or devices of the environment **100** may communicate in order to one or more of generate, train, or use a machine-learning model to process imaging data, among other activities.

[0031] As discussed in further detail below, the computing system(s) **102** may, one or more of, (i) generate, store, train, communicate with, or use a machine-learning model configured to process imaging data. The computing system(s) **102** may include a machine-learning model and/or instructions associated with the machine-learning model, e.g., instructions for generating a machine-learning model, training the machine-learning model, using the machine-learning model etc. The computing system(s) **102** may include instructions for retrieving data, adjusting data, e.g., based on the output of the machine-learning model, and/or operating a display of the user device(s) **112** to output generated responses to input, e.g., as adjusted based on the machine-learning model. The computing system(s) **102** may include training data, e.g., image data, and may include ground truth, e.g., (i) training whole slide images and (ii) training three-dimensional images to generate a navigable three-dimensional image.

[0032] As depicted in FIG. 1, computing system(s) **102** may include capturing module **104**. In various embodiments, capturing module **104** is configured to receive a plurality of whole slide images (WSI) associated with a tissue sample. The whole slide images and/or associated data may be gathered and/or compiled by the computing system **102** or using components separate from environment **100**. In examples, capturing module **104** may receive the whole slide images from medical computing system **116** via network **110**. Medical computing system **116** may be a user device associated with a medical institution, a medical imaging device, or the like. A medical imaging device implementing medical computing system **116** may include image processing system **102**, or image processing system **102** may be a separate component from medical computing system **116**. A plurality of images (e.g., digital or electronic image or a whole slide image (WSI)) may be received into electronic storage (e.g., cloud-based storage, hard disk, RAM, etc.) such as data store **114**. Further, and in various embodiments, capturing module **104** may receive patient data. In examples, patient data may include medical records, demographic information, medical predispositions, diagnoses and the like. Such patient data may be received by capturing module **104** from data store **114**, medical computing system **116**, user device **112**, or the like.

[0033] In example, such image data and patient data may be provided to one or more image processing machine-learning models. The one or more image processing machine-learning models may be implemented, generated, trained, or the like by machine-learning module **106**. The one or more image processing machine-learning models may be trained based on training data that includes historical/genuine/prior patient tissue images and/or simulated/synthetic image data, historical or simulated patient data, and/or the like. Synthetic image generation may use techniques described in U.S. application Ser. No. 17/645,197, which is incorporated herein by reference. The training data may be used to train the image processing machine-learning models by modifying one or more weights, layers, synapses, biases, and/or the like of the image processing machine-learning models, in accordance with a machine-learning algorithm, as discussed herein. Alternatively, or in addition, such image data may be used to generate a three-dimensional image.

[0034] Computing system(s) **102** may also include image generation module **107**. In various embodiments, image generation module **107** may be configured to generate a navigable three-

dimensional image of a tissue sample based on an output of the one or more machine-learning models. In various embodiments, image generation module **107** may also be configured to generate an interactive display that incorporates the navigable three-dimensional image. In various embodiments, the interactive display exists in an AR/VR environment.

[0035] In an embodiment, the AR headset may be used as a virtual microscope. Instead of using physical knobs and buttons to navigate the slide, the pathologist may use hand gestures, eye gestures, and so on as discussed herein, to do so, or be aided by an external device such as a computer mouse to navigate through the slide.

[0036] In examples, the interactive display enables a user to navigate aspects of the three-dimensional image (e.g., zoom in/out, rotate, flip, view a cross-section, “peel back” layers of the three-dimensional image to view interior aspects, and the like). Navigation could be controlled by various gestures. In one embodiment, a pathologist using an AR/VR environment while examining images may use hand gestures to navigate aspects of the three-dimensional image. The use of hand gestures is also useful for training purposes. For example, multiple pathologists may work on the same whole slide image in the same AR/VR environment, and the pathologist with the trainer role may point at a tissue region using hand gestures and describe it to the other pathologists. In this multi-user environment, virtual hands may be depicted based on hand-tracking of the users, so that others in the multi-user environment can see where fellow users are pointing.

[0037] In another embodiment, a pathologist using an AR/VR environment while examining digital pathology slides may use eye gestures to navigate the three-dimensional image. Eye gestures can be used to move slides, zoom in and out of regions on a slides, pan to certain regions of the slide based on gaze, auto-rotate of a 3D object. For example, blinking or double-blinking could be used to zoom into specific features of the 3D object. In another example, pathologists may navigate “up” and “down” the two-dimensional planes of a three-dimensional piece of tissue using eye gaze, where each plane corresponds to a WSI obtained from an individual glass slide.

[0038] In further examples, the interactive display that incorporates the navigable three-dimensional image in the AR/VR environment may be operable and/or configured to enable a user to navigate sample levels (e.g., tissue depths of the tissue sample associated with the image(s). Each level may be associated with a WSI.). In other various embodiments, image generation module **107** may be configured to generate a side-by-side display incorporating graphical representations of two or more images (e.g., whole slide images). In various additional embodiments, image generation module **107** may be configured to place a set of whole slide images in an order based an output of a machine-learning model and may be further configured to “stitch” the whole slide images together based on the ordering.

[0039] One embodiment of the invention relates to construction of 3D models using slicing and z-stacking. Various aspects of the present disclosure relate generally to computer-implemented techniques for image processing, such as whole slide images (WSI) obtained using medical imaging. Aspects disclosed herein may provide digital tools configured to conceptualize and reconstruct two-dimensional images (e.g., whole slide images), and to present them in an AR/VR environment. For example, aspects disclosed herein may “stitch” sequential images together to represent a three-dimensional (3D) structure that can be viewed, explored, and/or interrogated from different angles and/or different levels or depths (e.g., from each of those different angles).

[0040] Aspects disclosed herein may compare side-by-side and/or overlay spatially similar (e.g., based on depth) slides, or images of the same slide that has been treated in multiple steps (e.g., with a WSI image obtained at each step) for simultaneous or three-dimensional visualization. Stitching may refer herein to inferring orientation, inferring/filling 3D space (e.g., creating voxels) between single pairs of images that are known to be spatially similar (e.g., close together). In various embodiments, this may be done by using one or more machine-learning techniques and/or algorithms in sequence. For example, machine-learning techniques may be used to identify/recognize images with particular similarities (e.g., features, landmarks, or the like). An

algorithm may then be used to register the images (e.g., with an image registration algorithm such as linear, non-linear, rotational, rigid, and the like). Machine-learning techniques may then be used to infer/fill in the 3D space (e.g., create voxels). A mosaic may be a three-dimensional representation of a plurality of images and/or may refer to inferring the orientation of a larger set of images.

[0041] Aspects disclosed herein may provide solutions that allow for visualizing spatially similar sections of tissue by “stitching” images together to provide a single two-dimensional visualization and by rendering inferred three-dimensional visualizations from multiple two-dimensional whole slide images (WSIs). Users in an AR/VR environment may be able to “push or pull” on individual layers or portions of layers to transform them such that layers align more effectively. These solutions may also allow for inferring and rendering color of the three-dimensional visualizations.

[0042] As depicted in FIG. 1, computing system(s) **102** may also include transmission module **108**. In various embodiments, transmission module **107** may be configured to transmit the interactive display, the side-by-side display, and/or the generated navigable three-dimensional image to a user interface, such as of user device **112**. The user device **112** could be an AR/VR headset which allows a user to see an AR/VR environment while wearing the headset.

[0043] In further embodiments, transmission module **108** may be further configured to transmit the aforementioned to data store **114** (e.g., for storage or retention), or to medical computing system **116** (e.g., for storage, display, further processing, or the like).

[0044] As depicted in FIG. 1, environment **100** may also include electronic network **110**. In various embodiments, the electronic network **110** may be a wide area network (“WAN”), a local area network (“LAN”), personal area network (“PAN”), or the like. In some embodiments, electronic network **110** includes the Internet, and information and data provided between various systems occurs online. “Online” may mean connecting to or accessing source data or information from a location remote from other devices or networks coupled to the Internet. Alternatively, “online” may refer to connecting or accessing an electronic network (wired or wireless) via a mobile communications network or device. The Internet is a worldwide system of computer networks—a network of networks in which a party at one computer or other device connected to the network can obtain information from any other computer and communicate with parties of other computers or devices. The most widely used part of the Internet is the World Wide Web (often-abbreviated “WWW” or called “the Web”). A “website page” generally encompasses a location, data store, or the like that is, for example, hosted and/or operated by a computer system so as to be accessible online, and that may include data configured to cause a program such as a web browser to perform operations such as send, receive, or process data, generate a visual display and/or an interactive interface, or the like.

[0045] Although depicted as separate components in FIG. 1, it should be understood that a component or portion of a component in the environment **100** may, in some embodiments, be integrated with or incorporated into one or more other components. In another example, the computing system **102** may be integrated in a data storage system. The data storage system may be configured to communicate and/or receive/send data across electronic network **110** to other components of environment **100**. In some embodiments, operations or aspects of one or more of the components discussed above may be distributed amongst one or more other components. Any suitable arrangement and/or integration of the various systems and devices of the environment **100** may be used.

[0046] Further aspects of the computing system **102** and how a navigable three-dimensional image and/or an interactive display are generated are discussed in further detail in the methods below, with respect to FIGS. 2-3.

[0047] FIG. 2 depicts a flowchart of an exemplary method **200** for generating a navigable three-dimensional image of a tissue sample.

[0048] At step **205**, a plurality of whole slide images (WSI) are received (e.g., such as by capturing

module **104** as described with respect to FIG. **1**). The whole slide images may be associated with a tissue sample. For example, each whole slide image may be associated with a portion of the tissue sample imaged and/or sampled at a depth/level of the tissue sample. At step **210**, the plurality of whole slide images is provided to a machine-learning model. In various implementations, the machine-learning model had been trained, using one or more gathered prior patient and/or simulated sets of whole slide images and/or patient data, to identify one or more positional features within the plurality of whole slide images and output a plurality of relative positional relationships corresponding to each of the plurality of whole slide images.

[0049] Generally, an artificial intelligence or machine-learning model disclosed herein includes a set of variables, e.g., nodes, neurons, filters, etc., that are tuned, e.g., weighted or biased, to different values via the application of training data. In supervised learning, e.g., where a ground truth is known for the training data provided, training may proceed by feeding a sample of training data into a model with variables set at initialized values, e.g., at random, based on Gaussian noise, a pre-trained model, or the like. The output may be compared with the ground truth to determine an error, which may then be back-propagated through the model to adjust the values of the variable.

[0050] Training may be conducted in any suitable manner, e.g., in batches, and may include any suitable training methodology, e.g., stochastic or non-stochastic gradient descent, gradient boosting, random forest, etc. In some embodiments, a portion of the training data may be withheld during training and/or used to validate the trained machine-learning model, e.g., compare the output of the trained model with the ground truth for that portion of the training data to evaluate an accuracy of the trained model. The training of the machine-learning model may be configured to cause the machine-learning model to learn associations between image data and identify one or more positional features within the image data, such that the trained machine-learning model is configured to output a plurality of relative positional relationships corresponding to the image data (e.g., whole slide images).

[0051] In various embodiments, the variables of a machine-learning model may be interrelated in any suitable arrangement in order to generate the output. For example, in some embodiments, the machine-learning model may include image processing architecture that is configured to identify, isolate, and/or extract positional features in input image data. For example, the machine-learning model may include one or more convolutional neural network (“CNN”) configured to identify features in the whole slide images, and may include further architecture, e.g., a connected layer, neural network, etc., configured to determine a relative positional relationship between the identified features in order to generate a navigable three-dimensional image, or the like.

[0052] In some embodiments, the machine-learning or artificial intelligence model may include a Recurrent Neural Network (“RNN”). Generally, RNNs are a class of feed-forward neural networks that may be well adapted to processing a sequence of inputs. In some embodiments, the machine-learning model may include a Long Short Term Memory (“LSTM”) model and/or Sequence to Sequence (“Seq2Seq”) model. An LSTM model may be configured to generate an output from a sample that takes at least some previous samples and/or outputs into account. A Seq2Seq model may be configured to, for example, receive whole slide images as input, and generate an output. In some embodiments, the machine-learning model may include a transformer model and/or graph neural network (GNN) model. Such models may be configured to generate an output from input data.

[0053] In various embodiments, a method of training a machine-learning and/or artificial intelligence model to generate or provide a three-dimensional image and/or produce an interactive graphical user interface/virtual reality environment to display the tissue may include a step of receiving a plurality of images (e.g., electronic or digital images or whole slide images (WSIs)) into electronic or digital storage (e.g., cloud-based storage, hard disk, RAM, etc.). The images may be associated with a plurality of two-dimensional levels, dimensions, and/or views of a sample of a tissue of a patient. In some examples, the method may include a step of receiving additional data,

such as corresponding metadata (e.g., whether the slides are treated), patient data (e.g., diagnosis, disease progression, etc.), data input by a practitioner, and the like. In some examples, the images and/or additional data may be received for multiple points of time.

[0054] The method may include a step of training a machine-learning model or system to determine an order of the images and/or to produce a three-dimensional navigable image from the plurality of received images. The trained system may receive as input the received images and any additional data, and be trained to output an ordering or placement position of the received images to create a three-dimensional image, the three-dimensional image, and/a corresponding graphical user interface to navigate the three-dimensional image.

[0055] The system may be trained using weak supervision or strong supervision to identify regions of interest, biomarkers, or other landmarks and/or their dimensions that may help to determine the order of images, a depth level of the image, a relationship among the images, or the like. For example, the system may be trained using weak supervision, where a machine-learning model (e.g., multi-layer perceptron (MLP), convolutional neural network (CNN), Transformers, graph neural network, support vector machine (SVM), random forest, etc.) may utilize multiple instance learning (MIL) using weak labeling of the digital image or a collection of images. The labels of the training data may correspond to a positional label (e.g., order of the images, a depth of the sample that the image corresponds to, a stitching or mosaic label, coordinates, etc.) used to create the three-dimensional image. The trained model may predict a position arrangement, order, coordinates, etc. of the group of received images and/or of each individual image to co-register the images and create the three-dimensional image.

[0056] In some examples, instead of receiving labels for training, the system may receive a completed three-dimensional image or model, or a graphical user interface. The trained system may predict the three-dimensional image from the received images by, for example, identifying common salient regions among the images, analyzing their dimensions, analyzing color values, and determining their depths and/or placements (e.g., using edge detection, etc.).

[0057] For strongly supervised training, the image, the location of salient regions, an arrangement or order of the images, color values of each pixel, and/or a three-dimensional image to create may be received as input(s). Furthermore, information about whether the images include salient regions and/or certain diseases (e.g., whether they were malignant or benign) may also be received. For 2D images, e.g., whole slide images (WSI) in pathology, certain aspects (e.g., salient regions, disease information, and/or positional information) may be specified with pixel-level labeling, bounding box-based labeling or polygon-based labeling. While aspects disclosed herein are described in the context of stitching together WSIs, aspects disclosed herein may also be used to stitch together other types of images to create an interactive three-dimensional image, such as CT and MRI scans. For these other types of received images that may be more three-dimensional (e.g., CT and MRI scans), these aspects (e.g., salient regions, disease information, positional information) may be specified with voxel-level labeling, using a cuboid, etc. or using a parameterized representation allowing for sub-voxel-level labeling, such as parameterized curves or surfaces, or deformed template.

[0058] In some examples, the machine-learning model (e.g., R-CNN, Faster R-CNN, Selective Search, etc.) may be trained using bounding box or polygon-based supervision using bounding boxes or polygons that specify sub-regions of the received images that are salient, relevant, have certain color values for certain pixels, or as having a certain positional relationship (e.g., coordinates) to other sub-regions and/or images.

[0059] In some examples, the machine-learning model (e.g., Mask R-CNN, U-Net, Fully Convolutional Neural Network, Transformers, etc.) may be trained utilizing pixel-level or voxel-level labeling where individual pixels/voxels are identified as being salient, relevant, as having a certain color value, and/or as having a certain positional relationship to other pixels/voxels and/or images.

[0060] The machine-learning model may be trained to identify salient regions, biomarkers, color labels, etc. During training, other image processing techniques may be used, such as image segmentation or partitioning, using thresholding based on a variance of pixels in a tile to identify whether those pixels are foreground (and/or have a certain color value), using Otsu's method, comparing tile pixel values to a reference foreground distribution, etc. In some examples, the machine-learning model may be given input labels or segmentation masks describing salient regions or other relevant attributes or aspects. In some examples, the machine-learning model may be trained to extract a vector of features from each foreground tile to create a tile-level feature vector using a range of techniques such as hand-engineered features (e.g., scale invariant feature transform (SIFT) descriptors, oriented FAST and rotated BRIEF (ORB) descriptors, rotation invariant feature transform (RIFT) descriptors, speeded up robust features (SURF) descriptors, etc.), pre-trained CNN embeddings using supervised learning, pre-trained CNN embeddings using self-supervised learning techniques, pre-trained transformer neural network features, etc. The machine-learning model may learn to use the tile-level feature vectors in determining the three-dimensional image. For example, the machine-learning model may aggregate the tile-level feature vectors of each image (e.g., WSI) and classify the image as having certain attributes and/or corresponding to a certain depth of the tissue. The method may include a step of assigning, for each image, a label that indicates a positional relationship to the other images to be used for stitching.

[0061] The system may have been trained to infer three-dimensional voxels from two-dimensional pixel information obtained from the plurality of WSIs, which may represent multiple sequential, spatially similar slides, to determine, calculate, and/or create an image or model that can be navigated fully in three-dimensions, much like any computer-aided drawing interface. Specifically, the system may have been trained to create a three-dimensional model that is representative of a single stain while receiving slide inputs that are of a different stain or of a combination of stains (e.g., H&E and IHC).

[0062] The system may be trained to automatically co-register tissue for side-by-side navigation and/or an overlaid display. The system may be trained to automatically co-register the WSIs and/or coordinates of the sample on two or more WSIs, and determining the image may include automatically co-registering the WSIs and/or coordinates of the sample on two or more WSIs. The system may output (e.g., to memory and/or a display) the determined three-dimensional image and/or model.

[0063] The output may be configured to display a side-by-side display of multiple WSIs and/or slides based on the automatic co-registration of the coordinates of the tissue on two slides. Alternatively, or in addition thereto, the output may be configured to display an overlaid image of multiple WSIs as a single two-dimensional image. Displayed pixels may be based on an average of color information contained in the overlaid displayed pixels with differential staining. The display and/or output may provide a graphical user interface, which may include image editing and manipulation tools, such as white-balancing and color channel filtering, to highlight salient regions and/or features of interest. The salient regions may have been determined by an artificial intelligence and/or machine learning model or system. In various implementations, the display could be output to a heads-up display, such as a virtual/augmented/extended reality system/headset. The virtual/augmented/extended reality system may also be equipped with one or more components that enable navigation of the display in three-dimensional space. In other examples, the display may be output to an application installed on a user device (e.g., a web-based application, an application running locally on the user device, or the like). In further embodiments, image editing and manipulation tools may include two- and three-dimensional measurement tools (e.g., for linear, non-linear, polygon, and complex three-dimensional structures).

[0064] The system may include one or more computer vision algorithm(s) for rendering a more detailed or complete three-dimensional image. In addition, the system may include one or more computer vision algorithm(s) for comparing overlaid slides.

[0065] At step **215**, a navigable three-dimensional image may be generated based on the output relative positional relationships. At step **220**, an interactive display incorporating the navigable three-dimensional image may be generated inside an AR/VR environment. The AR/VR environment can be contoured to the room in which the user is viewing the three-dimensional rendering. For example, the rendering may be placed in the center of the room, or a predetermined minimum distance from each wall, if possible. The rendering may be placed a predetermined distance in front of the person who initiates the AR/VR view, based on the position of this person at the moment of initiating the AR/VR view, assuming that the rendering is also within a predetermined distance of any wall. The contours may be determined automatically via the system, wherein the three-dimensional rendering may be sized based on the determined contours of the room. For example, the rendering may be of a predetermined default size, which may be configurable by the user, but may be made larger or smaller to a certain predetermined extent, based on the size of the room. The AR/VR environment may also be integrated into a laboratory information system (LIS).

[0066] At step **225**, the interactive display may be transmitted to a user interface, the user interface being an AR/VR headset or comparable display. The AR/VR headset can utilize cameras that can record and track the eye movements of the user. Additionally, the AR/VR headset could also incorporate a microphone.

[0067] FIG. **3** depicts a flowchart of an exemplary method **300** for generating a navigable three-dimensional image of a tissue sample, according to an alternative embodiment. At step **305**, one or more whole slide images (WSI) may be received (e.g., such as by capturing module **104** as described with respect to FIG. **1**). The whole slide images may be associated with a tissue sample. For example, each whole slide image may be associated with a portion of the tissue sample imaged and/or sampled at a depth/z-level of the tissue sample. At step **310**, the plurality of whole slide images is provided to a machine-learning model. In various implementations, the machine-learning model had been trained, using one or more gathered prior patient and/or simulated sets of whole slide images and/or patient data, to identify one or more positional features within the plurality of whole slide images and output a sample level corresponding to each of the plurality of whole slide images. In examples, the sample level corresponds to a depth of the tissue sample.

[0068] A method of using a trained model or system may include a step of receiving a plurality of images (e.g., digital or electronic image or a whole slide image (WSI)) into electronic storage (e.g., cloud-based storage, hard disk, RAM, etc.)

[0069] The method may include a step of applying the trained machine-learning model to the plurality of images to co-register the images and/or otherwise determine a positional relationship, or, alternatively or in addition thereto, to create a three-dimensional image (e.g., based on the co-registration). For example, the method may include determining a depth level/z-level or order of the two-dimensional images, and overlaying the two-dimensional images according to the determined depth level. In some examples, the method may include determining a horizontal position of the two-dimensional images, and arranging two or more of the received two-dimensional images side-by-side to provide a continuous display of the sample. In some examples, the method may include determining a color label or value for various tiles and/or pixels of each image, and displaying a three-dimensional image that shows an average of the color values along the depths and/or levels for each pixel along a horizontal or vertical direction of the three-dimensional image.

[0070] In some examples, the method may include a step of applying the trained machine-learning model to allow a user in an AR/VR space to navigate the three-dimensional image to display the different levels, depths, views, and/or angles of the three-dimensional images based on the two-dimensional images. The user interface may be explored by a pathologist to better analyze the received images and sample.

[0071] At step **315**, each whole slide image may be ordered relative to the plurality of whole slide

images based on the sample level corresponding to each whole slide image. At step **320**, a navigable three-dimensional image may be generated using a stitching of the plurality of whole slide images based on the ordering. In examples, the interactive display is operable to navigate sample levels. At step **325**, an interactive display incorporating the navigable three-dimensional image is generated.

[0072] FIG. **4** depicts a flowchart of an exemplary method **400** for navigating a three-dimensional image in an AR/VR environment.

[0073] In method **400**, the three-dimensional image may be generated in an AR/VR environment, and may incorporate techniques discussed herein. At step **405**, a plurality of images may be received that correspond to or are associated with one or more WSIs. In other examples, using cameras on the AR/VR headset tracking the user's eye movement, the system can read a physical medical record file the user is looking at, query an LIS database, and retrieve relevant information such as patient health history, case history, and whole slide images. This additional information can be presented to the user alongside the three-dimensional image of the WSI.

[0074] In step **410**, the user can use gestures to control the three-dimensional image. In an example, a user could virtually pick up the slide and virtually move it with their hands. In an example, the system can track a user's eyes using the cameras of the headset to determine an eye gesture. The eye gestures could include winking, blinking, double-blinking, squinting, crossing of eyes, and other movements. The eye gesture could be used as an indication to zoom into features in the WSI. In another example, the user could cycle through different layers of the 3D object using an eye gesture. In another example, the user could use an eye gesture to lock onto a feature of the image, and use a second eye gestures to zoom into the feature. The specific use of an eye gesture could be pre-determined by the system or the user could create different settings for the eye gesture to take on a different meaning.

[0075] In step **420**, the AR/VR environment can be occupied by multiple participants. In this technique, the use of the AR/VR environment would be one on a collaboration tool or teaching tool. Multiple users could collectively view the same pathology slides using their own AR/VR interface. In step **430**, each pathologist may have their own view, or may choose to have their view synchronized across AR/VR interfaces.

[0076] In this shared AR/VR environment, users may have separate access and/or modification permissions. For example, one user may have the exclusive ability to change a view/slide/metadata display that is synchronized across multiple devices. In step **440**, a user might have the exclusive ability, depending on user access level, to annotate, apply virtual stains, or apply heat map layers. In an example, changes made by one user to the three-dimensional rendering may automatically propagate and update the three-dimensional rendering seen by other users substantially simultaneous with the change. In this example, when a user makes an annotation, other users in the viewing session, or that subsequently view the display, could be alerted via a beacon. This beacon might be the form of an alert or an arrow appearing in a screen. The beacon may appear in the corner of the display corresponding to the direction of the beacon. The users could then adjust their view to see the annotation, and the position of the beacon in the display may update dynamically.

[0077] As mentioned above, the system may be trained to automatically co-register tissue for side-by-side navigation and/or an overlaid display. The system may be trained to automatically co-register the WSIs and/or coordinates of the sample on two or more WSIs, and determining the image may include automatically co-registering the WSIs and/or coordinates of the sample on two or more WSIs. The system may output the determined three-dimensional image and/or model to be viewed in the AR/VR environment. The display and/or output may provide a graphical user interface in the AR/VR environment, which may include image editing and manipulation tools, such as white-balancing and color channel filtering, to highlight salient regions and/or features of interest. The salient regions may have been determined by an artificial intelligence and/or machine learning model or system.

[0078] During co-registration, the images of different levels of tissues may be moved around, rotated, and/or manipulated in order to allow for the identification of the match. These registration methods are mathematical algorithms that use shapes or images as input.

[0079] When multiple viewers are viewing the AR/VR space, there is a possibility that the user or users with the highest access privilege will accept the automatic co-registration of the images suggested by the system, which may be machine learning-based as discussed herein. In step **450**, these users might also suggest their own co-registration of the images which would perform a transformation on the image in order to change the stacking. In another example, a user with a lower access privilege might be able to suggest a different co-registration which the user with the higher access privilege could accept or reject. The user with the higher access privilege could also generate a voting poll for the user to vote on to determine which co-registration should be viewed by all viewers. Users with lower access levels could also annotate the images or suggest regions of interest (ROIs), and the grading of ROIs. A user with a predetermined access privilege could also directly “push or pull” on one or more z-layers simultaneously in the AR/VR environment, which may initiate a transformation of the affected layers, and which may cause a more accurate alignment of the layers of the z-stack.

[0080] As mentioned above, the system may include one or more computer vision algorithm(s) for rendering a more detailed or complete three-dimensional image. In addition, the system may include one or more computer vision algorithm(s) for comparing overlaid slides. In another embodiment, AR/VR may be used as a medium for visualizing electronic images of pathology slides that are generated via artificial intelligence (AI), machine learning (ML), deep learning, etc. AI may be a valuable tool in contextualizing information within the physical world. AI may be used as a mechanism of identification and visualization of characteristics within tissue. Only relevant parts may be extracted from electronic images of pathology slides. Regions of interest may be identified using AI/ML and automatically zoomed in according to the extent of the region of interest. Regions that are not of interest may be hidden from view, obscured, have the opacity lowered, etc. Furthermore, AI/ML may be used to remove floaters from tissues, floaters being pieces of tissue from another patient or small pieces of tissue from the same patient. Non-tissue items may also be removed from the images such as dust, or other contaminants. Regions that are automatically determined, via AI/ML, not to comprise tissue sample, e.g. background image, may also be removed.

[0081] As part of the AI/ML layer identification process, it may be inferred that there is a missing layer, and a corresponding alert may be generated to the user. The display in the AR/VR environment may be generated with a missing layer. However, when the machine learning system infers a layer based on an incomplete set of tissue layers, there is a chance that the inference is improper. In these cases, human intervention might be necessary to adjust weights of the system. In step **460**, a user with the highest access level could determine whether the correct layer ordering is being generated, and then propagate a changed layer to other users. In another example, a user with a lower access privilege might be able to suggest a different sample layer to produce which the user with the higher access privilege could accept or reject. The user with the higher access privilege could also generate a voting poll for the user to vote on to determine which layer should be viewed by all viewers.

[0082] As shown in FIG. 5, device **500** may include a central processing unit (CPU) **520**. CPU **520** may be any type of processor device including, for example, any type of special purpose or a general-purpose microprocessor device. As will be appreciated by persons skilled in the relevant art, CPU **520** also may be a single processor in a multi-core/multiprocessor system, such system operating alone, or in a cluster of computing devices operating in a cluster or server farm. CPU **520** may be connected to a data communication infrastructure **510**, for example a bus, message queue, network, or multi-core message-passing scheme.

[0083] Device **500** may also include a main memory **540**, for example, random access memory

(RAM), and also may include a secondary memory **530**. Secondary memory **530**, e.g. a read-only memory (ROM), may be, for example, a hard disk drive or a removable storage drive. Such a removable storage drive may comprise, for example, a floppy disk drive, a magnetic tape drive, an optical disk drive, a flash memory, or the like. The removable storage drive in this example reads from and/or writes to a removable storage unit in a well-known manner. The removable storage may comprise a floppy disk, magnetic tape, optical disk, etc., which is read by and written to by the removable storage drive. As will be appreciated by persons skilled in the relevant art, such a removable storage unit generally includes a computer usable storage medium having stored therein computer software and/or data.

[0084] In alternative implementations, secondary memory **530** may include similar means for allowing computer programs or other instructions to be loaded into device **500**. Examples of such means may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an EPROM or PROM) and associated socket, and other removable storage units and interfaces, which allow software and data to be transferred from a removable storage unit to device **500**.

[0085] Device **500** also may include a communications interface (“COM”) **560**. Communications interface **560** allows software and data to be transferred between device **500** and external devices. Communications interface **560** may include a modem, a network interface (such as an Ethernet card), a communications port, a PCMCIA slot and card, or the like. Software and data transferred via communications interface **560** may be in the form of signals, which may be electronic, electromagnetic, optical or other signals capable of being received by communications interface **560**. These signals may be provided to communications interface **560** via a communications path of device **500**, which may be implemented using, for example, wire or cable, fiber optics, a phone line, a cellular phone link, an RF link or other communications channels.

[0086] The hardware elements, operating systems, and programming languages of such equipment are conventional in nature, and it is presumed that those skilled in the art are adequately familiar therewith. Device **500** may also include input and output ports **550** to connect with input and output devices such as keyboards, mice, touchscreens, monitors, displays, etc. Of course, the various server functions may be implemented in a distributed fashion on a number of similar platforms, to distribute the processing load. Alternatively, the servers may be implemented by appropriate programming of one computer hardware platform.

[0087] Throughout this disclosure, references to components or modules generally refer to items that logically may be grouped together to perform a function or group of related functions. Like reference numerals are generally intended to refer to the same or similar components. Components and/or modules may be implemented in software, hardware, or a combination of software and/or hardware.

[0088] The tools, modules, and/or functions described above may be performed by one or more processors. “Storage” type media may include any or all of the tangible memory of the computers, processors or the like, or associated modules thereof, such as various semiconductor memories, tape drives, disk drives and the like, which may provide non-transitory storage at any time for software programming.

[0089] Software may be communicated through the Internet, a cloud service provider, or other telecommunication networks. For example, communications may enable loading software from one computer or processor into another. As used herein, unless restricted to non-transitory, tangible “storage” media, terms such as computer or machine “readable medium” refer to any medium that participates in providing instructions to a processor for execution.

[0090] One or more techniques presented herein may enable a user, to better interact with a digital image of a glass slide that may be presented on a screen, in a virtual reality environment, in an augmented reality environment, or via some other form of visual display. One or more techniques presented herein may enable a natural interaction closer to traditional microscopy with less fatigue

than using a mouse, keyboard, and/or other similar standard computer input devices.

[0091] The controllers disclosed herein may be comfortable for a user to control. The controllers disclosed herein may be implemented anywhere that digital healthcare is practiced, namely in hospitals, clinics, labs, and satellite or home offices. Standard technology may facilitate connections between input devices and computers (USB ports, Bluetooth (wireless), etc.) and may include customer drivers and software for programming, calibrating, and allowing inputs from the device to be received properly by a computer and visualization software.

[0092] Program aspects of the technology may be thought of as “products” or “articles of manufacture” typically in the form of executable code and/or associated data that is carried on or embodied in a type of machine-readable medium. “Storage” type media include any or all of the tangible memory of the computers, processors or the like, or associated modules thereof, such as various semiconductor memories, tape drives, disk drives and the like, which may provide non-transitory storage at any time for the software programming. All or portions of the software may at times be communicated through the Internet or various other telecommunication networks. Such communications, for example, may enable loading of the software from one computer or processor into another, for example, from a management server or host computer of the mobile communication network into the computer platform of a server and/or from a server to the mobile device. Thus, another type of media that may bear the software elements includes optical, electrical and electromagnetic waves, such as used across physical interfaces between local devices, through wired and optical landline networks and over various air-links. The physical elements that carry such waves, such as wired or wireless links, optical links, or the like, also may be considered as media bearing the software. As used herein, unless restricted to non-transitory, tangible “storage” media, terms such as computer or machine “readable medium” refer to any medium that participates in providing instructions to a processor for execution.

[0093] While the disclosed methods, devices, and systems are described with exemplary reference to transmitting data, it should be appreciated that the disclosed aspects may be applicable to any environment, such as a desktop or laptop computer, an automobile entertainment system, a home entertainment system, etc. Also, the disclosed aspects may be applicable to any type of Internet protocol.

[0094] It should be appreciated that in the above description of exemplary aspects of the invention, various features of the invention are sometimes grouped together in a single aspect, figure, or description thereof for the purpose of streamlining the disclosure and aiding in the understanding of one or more of the various inventive aspects. This method of disclosure, however, is not to be interpreted as reflecting an intention that the claimed invention requires more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive aspects lie in less than all features of a single foregoing disclosed aspect. Thus, the claims following the Detailed Description are hereby expressly incorporated into this Detailed Description, with each claim standing on its own as a separate aspect of this invention.

[0095] Furthermore, while some aspects described herein include some but not other features included in other aspects, combinations of features of different aspects are meant to be within the scope of the invention, and form different aspects, as would be understood by those skilled in the art. For example, in the following claims, any of the claimed aspects can be used in any combination.

[0096] Thus, while certain aspects have been described, those skilled in the art will recognize that other and further modifications may be made thereto without departing from the spirit of the invention, and it is intended to claim all such changes and modifications as falling within the scope of the invention. For example, functionality may be added or deleted from the block diagrams and operations may be interchanged among functional blocks. Operations may be added or deleted to methods described within the scope of the present invention.

[0097] The above disclosed subject matter is to be considered illustrative, and not restrictive, and

the appended claims are intended to cover all such modifications, enhancements, and other implementations, which fall within the true spirit and scope of the present disclosure. Thus, to the maximum extent allowed by law, the scope of the present disclosure is to be determined by the broadest permissible interpretation of the following claims and their equivalents, and shall not be restricted or limited by the foregoing detailed description. While various implementations of the disclosure have been described, it will be apparent to those of ordinary skill in the art that many more implementations are possible within the scope of the disclosure. Accordingly, the disclosure is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. A computer-implemented method for generating a navigable three-dimensional image of a tissue sample using augmented reality or virtual reality (AR/VR), the method comprising: receiving, at one or more processors, a plurality of medical images, the plurality of medical images being associated with at least one whole slide image (WSI) of the tissue sample; determining, by one or more processors, a layer level of each medical image of the plurality of medical images; determining, by one or more processors, for each medical image of the plurality of medical images, one or more regions of tissue and one or more regions of non-tissue; and stacking, by one or more processors, each medical image with one or more of the plurality of medical images, based on the layer level, to generate a three-dimensional rendering for display in an AR/VR environment.
2. The method of claim 1, further comprising: using one or more cameras on an AR/VR device to track a wearer's eye gestures.
3. The method of claim 2, wherein the wearer's eye gestures comprise blinking, double-blinking, winking, straining, squinting, crossing of eyes, and movement of the eyes.
4. The method of claim 3, further comprising: modifying one or more display settings of the three-dimensional rendering using the wearer's eye gestures, the one or more display settings comprising zooming into a feature on the three-dimensional rendering, cycling through the layer level of the three-dimensional rendering, or locking onto the feature on the three-dimensional rendering.
5. The method of claim 3, further comprising: modifying one or more display settings of a laboratory information system (LIS) using the wearer's eye gestures, the one or more display settings including reading a physical medical record file a user is looking at, querying an LIS database, and retrieving relevant information such as patient health history, case history, and whole slide images.
6. The method of claim 1, further comprising: receiving a user selection of a layer of the three-dimensional rendering; and generating a revised three-dimensional rendering, based on the user selection of the layer.
7. The method of claim 1, wherein a plurality of users have simultaneous access to the three-dimensional rendering, and each of the plurality of users are associated with a different access and/or modification permission.
8. The method of claim 7, wherein the different access and/or modification permission includes providing annotations of ROIs and/or providing suggestions of a user selection of a layer of the three-dimensional rendering.
9. The method of claim 1, further comprising: automatically determining one or more floaters and/or non-tissue items from each medical image of the plurality of medical images; and removing, from each medical image of the plurality of medical images, the floaters and/or non-tissue items.
10. The method of claim 1, wherein the AR/VR environment can be contoured to a room in which a user is viewing the three-dimensional rendering, wherein the contours are determined automatically, and wherein the three-dimensional rendering is sized based on the determined contours of the room.
11. A system for generating a navigable three-dimensional image of a tissue sample an augmented

reality or virtual reality (AR/VR) environment, the system comprising: a memory storing instructions and a processor operatively connected to the memory and configured to execute the instructions to perform operations comprising: receiving, at one or more processors, a plurality of medical images, the plurality of medical images being associated with at least one whole slide image (WSI) of the tissue sample; determining, by one or more processors, a layer level of each medical image of the plurality of medical images; determining, by one or more processors, for each medical image of the plurality of medical images, one or more regions of tissue and one or more regions of non-tissue; and stacking, by one or more processors, each medical image with one or more of the plurality of medical images, based on the layer level, to generate a three-dimensional rendering for display in an AR/VR environment.

12. The system of claim 11, further comprising: using one or more cameras on an AR/VR device to track a wearer's eye gestures, wherein the wearer's eye gestures comprise blinking, double-blinking, winking, straining, squinting, crossing of eyes, and movement of the eyes.

13. The system of claim 11, wherein a plurality of users have simultaneous access to the three-dimensional rendering, and each of the plurality of users are associated with a different access and/or modification permission.

14. The system of claim 13, wherein the different access and/or modification permission includes providing annotations of ROIs and/or providing suggestions of a user selection of a layer of the three-dimensional rendering.

15. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, perform operations including: receiving, at the one or more processors, a plurality of medical images, the plurality of medical images being associated with at least one whole slide image (WSI) of a tissue sample; determining, by one or more processors, a layer level of each medical image of the plurality of medical images; determining, by one or more processors, for each medical image of the plurality of medical images, one or more regions of tissue and one or more regions of non-tissue; and stacking, by one or more processors, each medical image with one or more of the plurality of medical images, based on the layer level, to generate a three-dimensional rendering for display in an AR/VR environment.

16. The computer-readable medium of claim 15, further comprising: using one or more cameras on the AR/VR device to track a wearer's eye gestures, wherein the wearer's eye gestures comprise blinking, winking, straining, squinting, crossing of eyes and sight movement.

17. The computer-readable medium of claim 16, further comprising: controlling one or more display settings of the three-dimensional rendering using the wearer's eye gestures, the one or more display settings including zooming into a feature on the three-dimensional rendering, cycling through the layer level of the three-dimensional rendering, or locking onto the feature on the three-dimensional rendering.

18. The computer-readable medium of claim 16, further comprising: controlling one or more display settings of a laboratory information system (LIS) using the wearer's eye gestures, the one or more display settings including reading a physical medical record file a user is looking at, querying an LIS database, and retrieving relevant information such as patient health history, case history, and whole slide images.

19. The computer-readable medium of claim 15, wherein a plurality of users have simultaneous access to the three-dimensional rendering, and each of the plurality of users are associated with a different access and/or modification permission.

20. The computer-readable medium of claim 19, wherein the different access and/or modification permission includes providing annotations of ROIs and/or providing suggestions of a user selection of a layer of the three-dimensional rendering.
